

Peace Umolu

Electronics & Embedded Systems Engineer

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PROFESSIONAL SUMMARY

Electronics & Embedded Systems Engineering student (Entering 3rd Year, First-Class Honors) with direct hands-on experience developing bare-metal firmware for **Analog Devices ADuCM410** (ARM Cortex-M33). Built a complete BMS test rig with FreeRTOS, 16-bit ADC with DMA, I2C/SPI drivers, and Python automated testing. Completed a **10-week intensive embedded systems program** on the ADuCM410 platform, producing 8 production-quality firmware drivers, complete OrCAD schematics, professional 4-layer PCB layout, and 200+ pages of documentation in Quarto. Maintains First-Class academic standing while working 36+ hours per week. Seeking to apply hardware/software co-design skills at **Analog Devices Limerick**.

TECHNICAL SKILLS

Languages	Embedded C, C++, Python (PyVISA, PySerial, Pytest), Java, Bash
Microcontrollers	ARM Cortex-M33 (ADuCM410), ARM Cortex-M4, STM32, ESP32
Protocols	I2C, SPI, CAN Bus, UART, MDIO, JESD204B (foundational)
Test Automation	Python (pytest, PyVISA, PySerial), MATLAB/Simulink
Tools	Keil MDK, IAR EWARM, OrCAD, KiCad, Git/GitHub, Linux, Quarto
Lab Instruments	DSO, Logic Analyzers (Saleae), Multimeters, Function Generators
Hardware	FreeRTOS, HAL Development, Register-Level Programming, DMA, ECC

EDUCATION

Technological University of the Shannon (TUS) *Limerick, Ireland*

- **B.Eng. (Honors) in Electronic Engineering with Computer Systems** *2024–Present*
- First-Class Honors. Achieved while managing 36+ hours/week in employment.
- *Coursework:* Microcontroller Systems, Embedded C, Digital Signal Processing, Analog & Digital Circuit Design, Data Communications, PCB Design & EDA.

IES Cabanyal *Valencia, Spain*

- **Advanced Certificate / Higher Certificate (NFQ Level 6)** *2013–2015*
- Focus: Industrial electrical installations, PLC programming, electronic troubleshooting, schematics.

TECHNICAL PROJECTS

ADI ADuCM410 10-Week Embedded Systems Portfolio

2026

- Developed comprehensive portfolio on ADuCM410QSPZ (ARM Cortex-M33) demonstrating full-stack embedded systems engineering.
- **Firmware:** 8 production-quality drivers including GPIO interrupts, UART CLI with Python automation, PWM motor control, 16-bit ADC with oversampling and temperature sensor, I2C/SPI with DMA, flash bootloader with trial run mode (as per ADI reference manual page 71-72), FreeRTOS with tasks/queues/semaphores, and power management with sleep modes.
- **Hardware:** Complete OrCAD system schematic (10+ pages) and professional 4-layer PCB layout.

- **Documentation:** 200+ page professional portfolio book in Quarto (HTML + PDF).
- *Portfolio:* peaum.github.io *GitHub:* github.com/peaum

Smart Single-Cell Battery Management System (BMS) & HIL Test Rig

2025–2026

- **Firmware (Embedded C):** Engineered multi-threaded FreeRTOS application monitoring Vcell, Iload, and temperature. Configured 16-bit ADC with DMA circular buffering (ADCCON, page 39; Table 183, page 135) for continuous background sampling without CPU overhead.
- **Communications:** Implemented non-blocking I2C and SPI telemetry drivers alongside real-time Coulomb Counting State of Charge (SoC) algorithm calibrated against OCV lookup tables.
- **Python Automation:** Built PyVISA test suite using pytest and pyserial to sweep bench supply voltages, log serial telemetry, and validate over-voltage fault trip execution within 50 ms.
- **Hardware Design:** Complete system schematic and PCB layout in OrCAD.

PROFESSIONAL EXPERIENCE

Acquired Brain Injury (ABI) Ireland

Tipperary, Ireland

– Rehabilitation Healthcare Assistant

July 2026–Present

- Deliver specialized person-centered rehabilitation support. Maintain rigorous health, safety, and compliance documentation under strict protocols.

Abbott Vascular

Clonmel, Ireland

– Technical Operative / Assembler

March 2025–Present

- Execute precision assembly and inspection of medical devices in high-compliance cleanroom environment. Strict adherence to Standard Operating Procedures (SOPs) and quality standards.

Access Healthcare

Ireland

– Healthcare Assistant (Agency)

2024–Present

- Provide high-demand care across healthcare facilities. Manage average of 36 hours weekly alongside full-time engineering studies.

Inforlandia

Valencia, Spain

– IT Support Technician & Frontend Developer

2017–2021

- Diagnosed hardware, board-level, and network faults. Configured software environments, drivers, and UIs.

B-Market / Jobelsa

Spain

– Electronic Technician & PLC Programmer

2015–2017

- Assembled and calibrated electronic tools. Programmed and configured PLCs for industrial production lines.

ADDITIONAL INFORMATION

Languages	Fluent in English and Spanish
Certifications	PHECC First Aid, Manual & Patient Handling, Safeguarding Adults, AMRIC Infection Control
Driving License	Full Clean B Irish Driving License
References	Available upon request